Agreement Attraction Meets Negative Concord:
Insights from Speeded Choice and Comprehension

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Agreement attraction happens when a verb erroneously agrees with an intervening *distractor* instead of the ***target*** (**The **key** to the cabinets are on the table*) [1, 2]. Number attraction has been widely observed with positive statements and positive intervenors [1, 2]. It is less clear what effect negation may have upon agreement (attraction) errors. We explore this question in Romanian, a negative concord language, where subject negative quantifiers must be in concord with negative verbs. Our study investigates whether negative intervenors can create number agreement attraction in negative contexts such as *Cheia de la **niciunele dintre cabinete nu** sunt pe masă,* lit. 'The key to none of the cabinets are not on the table'). Unlike nominal intervenors —which in Romanian are bare nouns, lack subject-like properties, and therefore induce only a small agreement attraction effect [3]—, negative quantifiers are subject-like, and thus we expect them to induce attraction. Under cue-based retrieval theories of agreement attraction [2, 4-5], participants could take the intervenor as the subject (*niciunele dintre cabinete* 'none of the cabinets'), an erroneous possibility reinforced by the negative verb. Under 'representational' theories of attraction [6], it is conceivable that (i) merely agreement features percolate to the head (*only agreement percolation,* i.e. the subject is *cheile* 'keys'), or (ii) that both agreement and negation features percolate (*agreement & negation percolation,* i.e. the subject is *niciunele dintre chei* 'none of the keys'). While under *agreement only percolation*, agreement is separate from negative features, under both *cue-based retrieval* and *agreement & negation percolation*, participants are susceptible to a *negative concord illusion* [7] with a negative subject. Moreover, under a *morpheme-item binding account* [8], agreement and/or negation could in principle erroneously attach to the wrong item/position.

To further shed light on the interplay between negative quantifiers and agreement attraction, and to explore what kind of subject illusions participants have in terms of comprehension, we ran a speeded forced choice continuation task [9], where participants had to choose between a 3rd singular and a 3rd plural verbal form, followed by a comprehension task where they had to specify the subject of the sentence [8, 9]. We tested 100 undergraduates (Romanian native speakers), but data collection is still in progress. There were 24 items with 6 conditions (see Table 1): MATCH (Number Match/ Mismatch) x NEGATION (Positive (No negation inside the DP)/Negative Head/Negative Intervenor). These were combined with 72 fillers. After each speeded critical item, participants were asked to choose the sentence subject from 8 choices (SG Head, Negative SG Head, SG Intervenor, Negative SG Intervenor, PL Head, Negative PL Head, PL Intervenor, Negative PL Intervenor). ***Results*** (see Table 2 & Fig 1). Agreement errors were overall quite low (under 15%). We compared accuracy rates across conditions using a generalized linear mixed-effects model with agreement accuracy as the dependent variable, the fixed effects Numbermatch (match = +0.5, mismatch = −0.5), **negation contrasts** using Helmert coding (ContrastA: neginterv = 1 vs neghead = −1, pos = 0; ContrastB: neginterv = 1 vs pos = −1, neghead = 0), and their interactions, random intercepts for participants and items, and corresponding random slopes. Overall, Number Mismatch yielded more error rates than Number Match. The main effect of Contrast A ($p < 0.01$) indicated that negative intervenors induce more errors than positive ones in negative polarity sentences. The interaction between Number Match and Contrast A was marginally significant ($p = 0.09$). Additionally, we found a significant effect for Contrast B between negative and positive intervenors ($p < .01$), and a significant interaction between Match and Contrast B ($p = 0.01$): positive intervenors induce a significant difference between match and mismatch, unlike negative ones.

Interestingly, the offline data reveals lower accuracy in comprehension vs. agreement (Fig.2). A regression model parallel to the one for agreement reveals significant effects for ContrastA and ContrastB in comprehension (Table 3). Moreover, we see a variety of errors, possibly in line with [8]: agreement percolation errors (highest number), agreement and negation percolation errors, cue retrieval errors, negation drop, a.o.

Overall, the online and offline data together suggest that the presence of negation on the intervenor makes processing more challenging: negative intervenors lead to higher overall agreement and comprehension errors but seem to prevent number agreement attraction.

Conditions exemplified	Example sentence
Number Match x Positive Head & Intervenor	Pisica de lângă fată nu au /are cat.the DE near girl not have.3pl/have.3sg
Number Match x Negative Intervenor	Pisica de lângă nicio fată nu au /are cat.the DE near no girl not have.3pl/have.3sg
Number Match x Negative Head	Nicio pisică de lângă fată nu au /are no cat DE near girl not have.3pl/have.3sg
Number Mismatch x Positive Head & Intervenor	Pisica de lângă fete nu au /are cat.the DE near girls not have.3pl/have.3sg
Number Mismatch x Negative Intervenor	Pisica de lângă niciunele dintre fetele nu au /are cat.the DE near none of girls.the not have.3pl/have.3sg
Number Mismatch x Negative Head	Nicio pisică de lângă fete nu au /are no cat DE near girls not have.3pl/have.3sg
Comprehension Question: *Cine nu are (ceva)?* 'Who does not have something?" *pisica* 'cat.the', *nicio pisică* 'no cat', *fata* 'girl', *nicio fată* 'no girl, *pisicile* 'cats.the', *niciunele dintre pisici* 'none of cats.the', *fetele* 'girls.the', *niciunele dintre* fete 'none of girls.the'	

Table 1. Example items

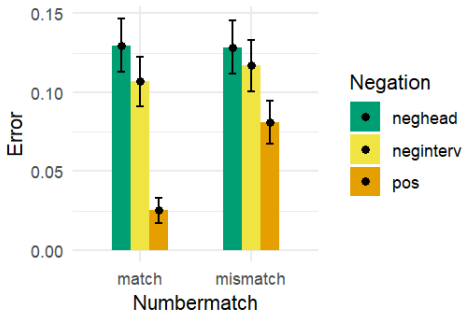

Figure 1. Agreement errors

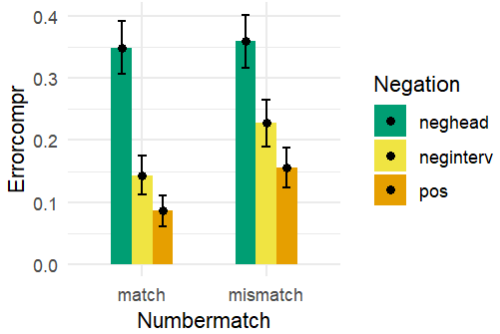

Figure 2. Comprehension errors

Parameter	Estimate	Std. error	z	p
Intercept	-2.886	0.172	-16.764	<.0001***
ContrastA (NegHead vs NegInterv)	-0.636	0.198	-3.204	0.001
Match	0.495	0.201	2.455	0.014
ContrastB (NegInterv vs Positive)	1.0315	0.2471	4.174	<.0001***
ContrastA:Match	0.414	0.251	1.652	0.098
Match:ContrastB	-0.706	0.296	-2.383	0.017

Table 2. Results of a generalized linear mixed effects model

Parameter	Estimate	Std. error	z	p
Intercept	-2.106	0.349	-6.016	<.0001***
ContrastA (NegHead vs NegInterv)	-1.245	0.258	-4.817	<.0001***
Match	0.595	0.317	1.877	0.06.
ContrastB (NegInterv vs Positive)	-1.245	0.258	-4.817	<.0001***
ContrastA:Match	0.511	0.326	1.564	0.117
Match:ContrastB	-0.373	0.359	-1.041	0.297

Table 3. Results of a generalized linear mixed effects model

References: [1] Bock & Miller, 1991. *Cognitive Psychology* [2] Wagers et al., 2009. *JML* [3] Bleotu & Dillon 2024[4] Dillon et al., 2013. *JML*. [4] Badeker & Kuminiak, 2007. *JML* [5] Slioussar, 2018. *JML* [6] Eberhard et al., 2005. *Psychol. Rev.* [7] Lacina, 2025. *Linguistics Vanguard*. [8] Keshev et al., 2025. *Topics in Cognitive Science,* [9] Staub, 2009. *JML* [10] Parker, 2019. *Glossa*.